A sharp stability threshold for Erdős Problem 848: admissible sets not contained in a single extremal class have density at most 1/50

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Abstract

Call $A \subseteq \{1, \dots, N\}$ *admissible* if $ab+1$ is not squarefree for all $a, b \in A$, including $a = b$. A. C. Li has determined the maximum size $f(N) = \lfloor (N+18)/25 \rfloor$ of an admissible set for every N, in a Lean-formalized proof; the extremal sets are the residue classes $n \equiv 7 \pmod{25}$ and $n \equiv 18 \pmod{25}$. Sawhney had previously proved a stability statement: there is an (unspecified) $\eta > 0$, implicitly $\eta \approx 0.0023$, such that for large N every admissible set of size at least $(1/25 - \eta)N$ lies inside one of these two classes; he left open the structure of admissible sets of smaller density. We close that gap in sharp form. We prove that for every $\varepsilon > 0$ and all $N \geq N_0(\varepsilon)$, every admissible $A \subseteq \{1, \dots, N\}$ which is *not* contained in a single one of the two classes satisfies $|A| \leq (1/50 + \varepsilon)N$, and that the constant $1/50$ is sharp, attained by the two residue-class constructions $\{n : \ n \equiv 7, 93 \pmod{100}\}$ and $\{n : \ n \equiv 43, 57 \pmod{100}\}$. The proof combines uniform analytic density bounds for the relevant squarefree-sieve events with two explicit finite computations, both described in the paper: the determination of the divisibility ("gluing") relation on residues modulo 900, and the maximization of the resulting weighted objective by 0–1 programming together with an exhaustive enumeration over the 439 compatible pairs of Galois-closed signature sets. The argument is self-contained: it uses neither Li's theorem nor Sawhney's, and it implies Sawhney's stability statement for every fixed $\eta < 1/50$; Li's theorem is quoted only for the exact extremal value $f(N)$ attained inside a single class. The argument is effective; an explicit (very weak) $N_0(\varepsilon)$ is recorded in Remark 1.6.

1 Introduction

1.1 The problem

For $N \in \mathbb{N}$ write $[N] = \{1, 2, \dots, N\}$.

Definition 1.1. A set $A \subseteq [N]$ is *admissible* if $ab+1$ is not squarefree for all $a, b \in A$, *including the diagonal case* $a = b$. We write

$$f(N) = \max\{|A| : \ A \subseteq [N] \text{ admissible}\}.$$

Erdős Problem 848 (Erdős–Sárközy) asks whether $f(N)$ is attained by the residue class $\{n \leq N : \ n \equiv 7 \pmod{25}\}$; see [1]. The diagonal convention $a = b$ is the one used in Sawhney's proof [4] and in Li's formalization [2]; it is essential below, since our argument extracts information from the non-squarefreeness of $a^2 + 1$ and gives nothing without it.

Two residue classes are admissible for trivial local reasons:

$$\mathcal{C}_N^7 = \{n \leq N : \ n \equiv 7 \ (25)\}, \qquad \mathcal{C}_N^{18} = \{n \leq N : \ n \equiv 18 \ (25)\},$$

because $7 \cdot 7 + 1 = 50$ and $18 \cdot 18 + 1 = 325$ are both divisible by 25, so that $25 \mid ab + 1$ whenever $a \equiv b \equiv 7$ (25) or $a \equiv b \equiv 18$ (25). (By contrast $7 \cdot 18 + 1 = 127$ is *not* divisible by 25: the two classes are not glued to each other by 25. This will matter.) We call these the two *extremal classes*.

1.2 Context: Li's theorem and Sawhney's stability theorem

Theorem 1.2 (A. C. Li [2])**.** *For every $N \geq 1$,*

$$f(N) = \big|\{n \leq N : \ n \equiv 7 \pmod{25}\}\big| = \Big\lfloor \frac{N+18}{25} \Big\rfloor.$$

Theorem 1.2 is proved in [2] by an exact prefix colouring for $N \leq 5 \cdot 10^6$ combined with a Hall-defect sieve argument for larger N, and is formalized in Lean 4 in the accompanying repository. It is quoted here for context only: the proof of Theorem 1.4 neither uses, reproves, nor modifies it. (Section 9 records an independent source rebuild and external kernel replay of that formalization, performed by the present author; that work verifies Li's theorem and is logically separate from the theorem proved here.)

Theorem 1.3 (Sawhney [4])**.** *There is $\eta > 0$ such that for all sufficiently large N, every admissible $A \subseteq [N]$ with $|A| \geq (1/25 - \eta)N$ satisfies $A \subseteq \mathcal{C}_N^7$ or $A \subseteq \mathcal{C}_N^{18}$.*

Sawhney does not specify η; his four case bounds are $0.0377N$, $0.0358N$, $0.0336N$, $0.0294N$ (each $+o(N)$), so that $\eta \approx 0.0023$ is implicit. He explicitly leaves open "a strong structural classification of sets A with size δN". Thus before the present note, the structure of admissible sets was understood only in the top sliver of densities, $|A|/N \in [1/25 - 0.0023,\ 1/25]$, and in the whole middle range below it only non-sharp bounds were known (see Remark 1.7).

1.3 Main theorem

Theorem 1.4. *For every $\varepsilon > 0$ there is $N_0(\varepsilon)$ such that the following holds for all $N \geq N_0(\varepsilon)$. If $A \subseteq [N]$ is admissible and A is* not *contained in $\mathcal{C}_N^7$ and* not *contained in $\mathcal{C}_N^{18}$, then*

$$|A| \ \leq \ \Big(\frac{1}{50} + \varepsilon\Big)N.$$

Equivalently: every admissible set with more than $(1/50 + \varepsilon)N$ elements lies inside one of the two extremal classes. The constant $1/50$ is sharp: it is attained, within the mod-900 type framework of Section 6 and up to $O(1)$ errors, by exactly two residue-class constructions,

$$A_1 = \{n \leq N : \ n \equiv 7 \text{ or } 93 \pmod{100}\}, \qquad A_2 = \{n \leq N : \ n \equiv 43 \text{ or } 57 \pmod{100}\},$$

both of which are admissible, meet both extremal classes, and have $|A_i| = N/50 + O(1)$.

Combined with Theorem 1.2, the picture for large N is therefore complete up to ε: an admissible set either sits inside one extremal class (where its size is at most $\lfloor (N+18)/25 \rfloor$, with equality for the class itself), or it has density at most $1/50 + \varepsilon$, and the latter bound is attained.

Remark 1.5 (Effectivity). The density bounds of Section 4 are stated in an ε–T–N_0 form in which the truncation parameter $T = T(\varepsilon)$ and then N_0 are chosen but not tracked. This is a matter of convenience rather than of principle: the tail estimate of Lemma 4.1 is elementary (it uses only $q^2 \leq ab + 1 \leq N^2 + 1$ for the prime squares q^2 in question), and the remaining ingredients are the finite congruence computations of Sections 3 and 6. The argument is

therefore effective, and Remark 1.6 below extracts an explicit (extremely weak) $N_0(\varepsilon)$ by tracking $T(\varepsilon)$, the tail sum over primes $q > T$, and the $O_T(1)$ CRT error. We stress that the proof of Theorem 1.4 is logically independent of [4]: no estimate of Sawhney's is used, and no ineffectivity is inherited from it. Li's theorem [2] is likewise not used in the proof of Theorem 1.4; it is quoted only for the exact extremal value $f(N) = \lfloor (N+18)/25 \rfloor$ inside a single extremal class, in the surrounding discussion.

Remark 1.6 (An explicit, very weak N_0). The ineffectivity above is only a matter of convenience. Tracking the choices in Section 4 with deliberately crude bounds: truncating each Euler product at T moves the constants K_2' and J' by at most $8/T$ (each product tail is at most $2\sum_{q>T} q^{-2} \le 2/T$), so $T = \lceil 24/\varepsilon \rceil$ makes the density error at most $\varepsilon/3$ and also gives $3/T \le \varepsilon/3$ for the main term of Lemma 4.1; the remaining error terms are the $3\pi(N)$ of that lemma and the CRT error, which is at most the number $Q_T = \prod_{7 \le q \le T} q^2 \le e^{2T \log T}$ of residue classes involved. By Rosser's bound $\pi(x) < 1.25506\, x/\log x$, the condition $\log N \ge 20332/\varepsilon$ makes $3\pi(N) \le \varepsilon N/(6M)$; it also implies $Q_T \le \varepsilon N/(6M)$ provided $10^{-100} \le \varepsilon \le 1/100$, since $\log Q_T \le 2T \log T$ with $T = \lceil 24/\varepsilon \rceil$ and $2T \log T \le 20332/\varepsilon - \log(6M/\varepsilon)$ throughout that range (for smaller ε the same tracking gives $\log N_0(\varepsilon) = O(\varepsilon^{-1} \log(1/\varepsilon))$). Hence one may take

$$N_0(\varepsilon) = \exp\bigl(20332/\varepsilon\bigr) \qquad (10^{-100} \le \varepsilon \le 1/100),$$

and in particular $N_0(1/100) \le 10^{883008}$. These values are of course far from optimal; we record them only to show that an explicit threshold can be extracted from the argument.

Remark 1.7 (Prior work and what is new). The observation that the residues 7 and 93 modulo 100 are compatible "through 4", because $7 \cdot 93 + 1 = 652 = 4 \cdot 163$, is due to **AstroCat93** [5], whose verification write-up also isolates a sub-case of limiting density 0.02; its overall bound is ≈ 0.0282. That work does not show that density $1/50$ is *attained* by an admissible set, and does not establish $1/50$ as the global stability threshold. What is new here is (i) the proof that the full configuration $\{7, 93\}$ (mod 100) (and its mirror $\{43, 57\}$ (mod 100)) is admissible and attains density exactly $1/50$, and (ii) the matching sharp upper bound of Theorem 1.4, obtained by reducing all mixed configurations to the single numerical inequality $72K_2' + 828J' < 18$ of Section 6. Other community bounds in circulation are weaker: ≈ 0.0276 [7] and an existential $\eta = 1/2000$ in the Lean work of [8]. Neither the $1/50$ threshold nor the two constructions are stated in any of the works cited in this remark.

2 The extremal configurations

Proposition 2.1. *The sets* $A_1 = \{n \le N : n \equiv 7, 93 \ (100)\}$ *and* $A_2 = \{n \le N : n \equiv 43, 57 \ (100)\}$ *are admissible, each meets both extremal classes, and* $|A_i| = N/50 + O(1)$.

Proof. Diagonal and within-residue pairs. If $a \equiv b \equiv 7$ (100) then $ab + 1 \equiv 50$ (100), so $25 \mid ab + 1$; in particular $7^2 + 1 = 50$. If $a \equiv b \equiv 93$ (100) then again $ab + 1 \equiv 93 \cdot 93 + 1 = 8650 \equiv 50$ (100), and $8650 = 2 \cdot 5^2 \cdot 173$, so $25 \mid ab + 1$. Cross pairs. If $a \equiv 7$ and $b \equiv 93$ (100) then $ab + 1 \equiv 7 \cdot 93 + 1 = 652 \equiv 52$ (100), so $4 \mid ab + 1$ (indeed $652 = 2^2 \cdot 163$). In all cases $ab + 1$ is divisible by 25 or by 4, hence not squarefree, so A_1 is admissible.

For A_2: $43^2 + 1 = 1850 = 2 \cdot 5^2 \cdot 37$ and $57^2 + 1 = 3250 = 2 \cdot 5^3 \cdot 13$ are both divisible by 25, so within-residue pairs give $ab + 1 \equiv 50$ (100); and $43 \cdot 57 + 1 = 2452 \equiv 52$ (100) is divisible by 4. So A_2 is admissible.

Finally $7 \equiv 7$ and $93 \equiv 18$ (25), while $43 \equiv 18$ and $57 \equiv 7$ (25); so each A_i meets both extremal classes and is contained in neither. Each A_i consists of two residue classes mod 100, whence $|A_i| = 2N/100 + O(1) = N/50 + O(1)$. □

Proposition 2.2 (Finite check)**.** *Among all pairs* $\{u, v\}$ *of residues mod* 100 *with* $u \equiv 7$, $v \equiv 18 \pmod{25}$ *(or vice versa), the only pairs for which* $\{n \le N : \ n \equiv u \text{ or } v \ (100)\}$ *is admissible are* $\{7, 93\}$ *and* $\{43, 57\}$.

Proof. There are $4 \cdot 4 = 16$ such pairs, and for each the three conditions ($u^2 + 1$, $v^2 + 1$, $uv+1$ not squarefree by a local obstruction at 4, 9 or 25) are decided by a direct computation modulo 900. The enumeration was carried out by the author and independently by a referee whose report and code are distributed with [6], and also checked against all actual integer pairs up to 20000: for each of the remaining fourteen pairs that check exhibits explicit elements a, b with $ab + 1$ squarefree, so those class pairs are genuinely inadmissible and not merely unobstructed. □

Remark 2.3 (The cross-class check by hand). The cross-class part of the computation above can also be done by hand. Let $u \equiv 7 \ (25)$ and $v \equiv 18 \ (25)$. Then $uv + 1 \equiv 7 \cdot 18 + 1 = 127 \equiv 2 \ (25)$, so $25 \nmid uv + 1$, and the only prime square that can obstruct $uv + 1$ for every choice of integers u, v in these classes is 4. Hence we need $uv \equiv 3 \ (4)$, i.e. $\{u \bmod 4, \ v \bmod 4\} = \{1, 3\}$. Modulo 100 the relevant residues are $u \in \{7, 32, 57, 82\}$, with $u \equiv 3, 0, 1, 2 \ (4)$ respectively, and $v \in \{18, 43, 68, 93\}$, with $v \equiv 2, 3, 0, 1 \ (4)$ respectively. The only pairs with $uv \equiv 3 \ (4)$ are therefore $(u, v) = (7, 93)$ and $(u, v) = (57, 43)$, which are exactly the pairs underlying A_1 and A_2.

3 Types modulo 900 and gluing

Throughout, set

$$M = 900 = 4 \cdot 9 \cdot 25.$$

The *type* of an integer n is its residue $n \bmod M$. Let

$$\mathcal{D} = \{p \text{ prime} : \ p \equiv 1 \pmod 4, \ p \ge 13\}.$$

For integers a, b define the two sieve events

$$D(a) : \ \exists p \in \mathcal{D}, \ p^2 \mid a^2 + 1; \qquad P_b(a) : \ \exists q \text{ prime}, \ q \ge 7, \ q^2 \mid ab + 1.$$

Lemma 3.1 (Local facts; finite verification)**.** *(F1) For every* n, n^2+1 *is divisible by neither* 4 *nor* 9*; it is divisible by* 25 *if and only if* $n \equiv 7$ *or* $18 \pmod{25}$*; no prime* $p \equiv 3 \pmod 4$ *divides* $n^2 + 1$*; and for* $p \equiv 1 \pmod 4$ *the congruence* $x^2 \equiv -1 \pmod{p^2}$ *has exactly two roots. Consequently, if* $n \bmod 25 \notin \{7, 18\}$*, then*

$$n^2 + 1 \text{ is not squarefree} \iff D(n).$$

(F2) Whether 4, 9 *or* 25 *divides* $ab + 1$ *depends only on the types of* a *and* b*. Call a pair of types* (s, t) glued *if* $4 \mid st + 1$ *or* $9 \mid st + 1$ *or* $25 \mid st + 1$*. If* (s, t) *is not glued, then for* a *of type* s *and* b *of type* t*,*

$$ab + 1 \text{ is not squarefree} \iff P_b(a),$$

and necessarily $q \nmid b$ *for the witnessing prime* q*.*

(F3) Let $C \subseteq \mathbb{Z}/M\mathbb{Z}$ *be the set of* 72 *types* s *with* $s \equiv 7$ *or* $18 \pmod{25}$*. Every* $s \in C$ *is glued to itself and to every type in the same class mod* 25*. A type outside* C *is never glued to itself. A type in* C *and a type outside* C *can be glued only through* 4 *or* 9*; likewise a* 7*-class type and an* 18*-class type can be glued only through* 4 *or* 9*.*

Proof. (F1) Squares mod 4 lie in $\{0,1\}$ and squares mod 9 in $\{0,1,4,7\}$, so $n^2+1 \not\equiv 0$ mod 4 or 9. The roots of $x^2 \equiv -1 \pmod{25}$ are $x \equiv 7, 18$. If $p \equiv 3 \pmod 4$ then -1 is a non-residue mod p. If $p \equiv 1 \pmod 4$, Hensel lifting gives exactly two roots mod p^2. Hence, for n with $n \bmod 25 \notin \{7, 18\}$, the only primes whose square can divide n^2+1 are $p \equiv 1 \pmod 4$ with $p \neq 5$, i.e. $p \in \mathcal{D}$.

(F2) The first assertion is clear since $4, 9, 25 \mid M$. If (s,t) is not glued, then $4, 9, 25 \nmid ab+1$; since $2^2 = 4$, $3^2 = 9$, $5^2 = 25$, any prime square dividing $ab+1$ has prime $q \geq 7$. If $q \mid b$ and $q^2 \mid ab+1$ then $q \mid 1$, a contradiction.

(F3) If $s \equiv t \equiv 7 \pmod{25}$ then $st+1 \equiv 50 \equiv 0 \pmod{25}$, and similarly $18 \cdot 18 + 1 = 325 \equiv 0 \pmod{25}$; this covers self-gluing and same-class gluing. If $s \notin C$ then s^2+1 is divisible by none of $4, 9, 25$ by (F1). Finally, if $25 \mid st+1$ with $s \equiv 7 \pmod{25}$ then $t \equiv -7^{-1} \equiv -18 \equiv 7 \pmod{25}$ (as $7 \cdot 18 = 126 \equiv 1$), so t is in the same class; the same computation with 7 and 18 interchanged gives the remaining statement. □

The local facts were verified computationally as well: (F1) for all primes $p < 2000$, (F2) on 20,000 random pairs, and (F3) on all 900×900 type pairs (see Section 9).

4 Uniform density bounds

All bounds in this section are uniform in the "pivot" $b \leq N$, which is what allows us to choose the witness b depending on the set A.

Lemma 4.1 (Truncation)**.** *Let $T \geq 7$ and let E be either of the events P_b, $D \wedge P_b$. For a fixed residue s mod M, the count $\#\{a \leq N : a \equiv s\ (M),\ E(a)\}$ differs from the corresponding count with all witnessing primes restricted to $q \leq T$ by at most*

$$3 \sum_{T < q \leq N} \Big(\frac{N}{Mq^2} + 1 \Big) \ \leq \ \frac{3N}{MT} + 3\pi(N),$$

and the truncated count is an exact CRT count $\mathrm{dens}_T \cdot N/M + O_T(1)$*. Only primes $q \leq N$ need be considered, since $q^2 \leq ab+1 \leq N^2+1$.*

Indeed, if $E(a)$ holds while its T-truncation fails, some prime $q > T$ witnesses $D(a)$ or $P_b(a)$: the first alternative puts a in two residue classes mod q^2 when $q \in \mathcal{D}$ (and is impossible for $q \notin \mathcal{D}$), the second in a single class, so at most three classes mod q^2 in all. Since $q \geq 7$ is coprime to M, each such class meets the fixed class s mod M in a single class mod Mq^2, which contains at most $N/(Mq^2)+1$ integers in $[N]$.

Given $\varepsilon > 0$ we first choose $T = T(\varepsilon)$ so that the truncated CRT densities are within $\varepsilon/2$ of their limits and $3/T < \varepsilon/4$, and then N_0 so that $3\pi(N) + O_T(1) < \varepsilon N/(4M)$ for $N \geq N_0$. All constants below are limits of truncated products, and only upper bounds on them are used.

Lemma 4.2 (The K_2' cap)**.** *For every $\varepsilon > 0$ there is N_0 such that for all $N \geq N_0$, all $b \leq N$ and all residues s mod M,*

$$\#\{a \leq N : a \equiv s\ (M),\ P_b(a)\} \ \leq \ (K_2' + \varepsilon)\frac{N}{M}, \qquad K_2' = 1 - \prod_{q \geq 7} \big(1 - q^{-2}\big) = 1 - \frac{9.375}{\pi^2} \approx 0.0501139.$$

Proof. By (F2) a witnessing prime satisfies $q \geq 7$ and $q \nmid b$; for such q the condition $q^2 \mid ab+1$ puts a in a single residue class mod q^2, and the classes for distinct q are independent by CRT. Hence the truncated density of P_b is $1 - \prod_{7 \leq q \leq T,\, q \nmid b}(1-q^{-2}) \leq 1 - \prod_{7 \leq q \leq T}(1-q^{-2})$, and letting $T \to \infty$ in the majorant gives K_2'. Finally $\prod_{q \geq 7}(1-q^{-2}) = \big(6/\pi^2\big)/\big(\frac{3}{4} \cdot \frac{8}{9} \cdot \frac{24}{25}\big) = 9.375/\pi^2$, since $\frac{3}{4} \cdot \frac{8}{9} \cdot \frac{24}{25} = 0.64$. Apply Lemma 4.1. □

Lemma 4.3 (The J' cap)**.** *Put*

$$A_{\mathcal{D}} = \prod_{p\in\mathcal{D}}\bigl(1-2p^{-2}\bigr), \quad B_{\mathcal{D}} = \prod_{p\in\mathcal{D}}\bigl(1-p^{-2}\bigr), \quad Q = \prod_{\substack{p\equiv 3\ (4)\\ p\geq 7}}\bigl(1-p^{-2}\bigr),$$

$$K_1 = 1 - A_{\mathcal{D}} \approx 0.0273465, \qquad J' = K_1 - Q\,\bigl(B_{\mathcal{D}} - A_{\mathcal{D}}\bigr) \approx 0.0142450.$$

Then for every $\varepsilon > 0$ there is N_0 such that for all $N \geq N_0$, all $b \leq N$ and all residues s mod M with $s \notin C$,

$$\#\{a \leq N :\ a \equiv s\ (M),\ D(a) \wedge P_b(a)\} \ \leq\ (J' + \varepsilon)\frac{N}{M}.$$

Proof. By inclusion–exclusion, for the truncated events,

$$\operatorname{dens}\bigl(D \wedge P_b\bigr) = 1 - \prod_{\ell} Y_\ell - \prod_{\ell} X_\ell + \prod_{\ell} Z_\ell,$$

where X_ℓ, Y_ℓ, Z_ℓ are the local densities at ℓ of $\neg P_b$, $\neg D$ and $\neg D \wedge \neg P_b$ respectively, and $Z_\ell \leq X_\ell$ at every prime. Explicitly: for $\ell \in \mathcal{D}$ the set $D_\ell = \{\pm r\}$ with $r^2 \equiv -1\ (\ell^2)$ has two elements mod ℓ^2, and $P_\ell = \{-b^{-1} \bmod \ell^2\}$ is a single class when $\ell \nmid b$, empty when $\ell \mid b$; for $\ell \geq 7$, $\ell \notin \mathcal{D}$ (i.e. $\ell \equiv 3\ (4)$, $\ell \geq 7$) the event D is locally empty.

The density is a product of positive local factors, so the dependence on b may be analysed one prime at a time, all other factors held fixed, and the signs of the two possible changes do not depend on the other factors:

- If $\ell \mid b$ then $P_\ell = \emptyset$, and the change in the density relative to $\ell \nmid b$ is $(Z' - X')/\ell^2 \leq 0$ if P_ℓ would lie outside D_ℓ, and $-X'/\ell^2 \leq 0$ if it would lie inside, where X', Z' denote the products of the remaining local factors. So $\gcd(b, \ell) = 1$ is the worse (larger) case.
- If $\ell \in \mathcal{D}$ and $P_\ell \subset D_\ell$ (equivalently $\ell^2 \mid b^2 + 1$), then Z_ℓ rises from $1 - 3/\ell^2$ to $1 - 2/\ell^2$, which raises the density. So coincidence is the worse case.

Taking the worst case at every prime simultaneously therefore majorizes every b. The constant J' so obtained is a formal majorant: the worse alternative is taken independently at each prime, and no single integer b need realize all of them at once (indeed $\ell^2 \mid b^2 + 1$ can hold for only finitely many ℓ, since $\ell^2 \leq b^2 + 1$); the bound is valid for every b regardless. In that worst case $\prod Y = A_{\mathcal{D}}$, $\prod X = Q\,B_{\mathcal{D}} = 1 - K_2'$ and $\prod Z = Q\,A_{\mathcal{D}}$, giving

$$1 - A_{\mathcal{D}} - QB_{\mathcal{D}} + QA_{\mathcal{D}} = K_1 - Q\bigl(B_{\mathcal{D}} - A_{\mathcal{D}}\bigr) = J'.$$

Finally $s \notin C$ guarantees, by (F1), that non-squarefreeness of $a^2 + 1$ is exactly the event $D(a)$, so the bound is applicable. Apply Lemma 4.1. □

The product formula underlying Lemma 4.3 was checked exactly against full CRT-period counts on the prime sets $\{7, 13\}$, $\{7, 13, 17\}$, $\{11, 13, 17\}$ and $\{7, 11, 13\}$, in each case with the maximum over b equal to the predicted worst case. Empirically, for $a \leq 10^7$ and pivots $b \in \{1,\ 30030,\ 9699690,\ 9999991,\ 99,\ 3141,\ 33459226\}$ (exhibiting 0 to 3 coincidences $\ell^2 \mid b^2+1$), all observed densities matched the exact per-b predictions ($|z| \leq 0.4$), the largest observed $D \wedge P_b$ density being $0.01120 < J'$. Pivots with ≥ 4 coincidences were not tested.

Constant	Definition	Value / enclosure
$A_{\mathcal{D}}$	$\prod_{p\in\mathcal{D}}(1-2p^{-2})$	0.9726535060
$B_{\mathcal{D}}$	$\prod_{p\in\mathcal{D}}(1-p^{-2})$	0.9862566747
Q	$\prod_{p\equiv 3(4),\,p\geq 7}(1-p^{-2})$	0.9631226050
K_1	$1-A_{\mathcal{D}}$	≈ 0.0273465, [0.02734649, 0.02734654]
K_2'	$1-\prod_{q\geq 7}(1-q^{-2}) = 1-9.375/\pi^2$	≈ 0.0501139, [0.05011390, 0.05011395]
J'	$K_1 - Q(B_{\mathcal{D}}-A_{\mathcal{D}})$	≈ 0.0142450, [0.01424497, 0.01424500]
generic b	density of $D\wedge P_b$, no coincidences	≈ 0.0012774

Table 1: The density constants. Enclosures were obtained from partial products over primes up to $P = 5\cdot 10^7$ with rigorous tails (using $\sum_{n>P} n^{-2} < 1/(P-1)$); the author's independent computation used $P = 2\cdot 10^7$ and agrees with these enclosures to within $2\cdot 10^{-7}$.

5 The per-type bound

Let $A\subseteq[N]$ be admissible and let $S\subseteq\mathbb{Z}/M\mathbb{Z}$ be the set of types occurring in A. For $s\in S$ define the weight

$$f(s,S) = \begin{cases} 1, & s\in C \text{ and } s \text{ is glued to every } t\in S,\\ K_2', & s\in C \text{ and some } t\in S \text{ is not glued to } s,\\ J', & s\notin C. \end{cases}$$

Proposition 5.1. *For every $\varepsilon>0$ there is $N_0(\varepsilon)$ such that for all $N\geq N_0$ and all admissible $A\subseteq[N]$ with type set S,*

$$|A| \;\leq\; \frac{N}{M}\sum_{s\in S} f(s,S) \;+\; \varepsilon N \;+\; O(M).$$

Proof. Write $A_s = \{a\in A : a\equiv s\ (M)\}$, so $|A| = \sum_{s\in S}|A_s|$. There are at most M summands, so it suffices to bound each $|A_s|$ by $f(s,S)N/M + \varepsilon N/M + O(1)$.

If $s\in C$ is glued to every $t\in S$ we use the trivial bound $|A_s|\leq N/M+1$.

If $s\in C$ and some $t\in S$ is not glued to s, pick $b\in A$ of type t. By (F3) a C-type is glued to itself, so $t\neq s$ and hence $b\notin A_s$. For every $a\in A_s$, $ab+1$ is not squarefree and (s,t) is not glued, so $P_b(a)$ holds by (F2). Lemma 4.2 gives $|A_s|\leq (K_2'+\varepsilon)N/M$.

If $s\notin C$ and $|A_s|\geq 2$, pick $b\in A_s$. By (F3), (s,s) is not glued, so every $a\in A_s\setminus\{b\}$ satisfies $P_b(a)$ by (F2); and a^2+1 is not squarefree (diagonal admissibility) with $s\notin C$, so $D(a)$ holds by (F1). Lemma 4.3 gives $|A_s|\leq 1+(J'+\varepsilon)N/M$. If $|A_s|\leq 1$ its contribution is at most 1, which is absorbed in the $O(M)$ term. □

Remark 5.2 (The diagonal condition is essential). The argument just given uses the diagonal case $a=b$ of admissibility in an essential way: it is what forces $D(a)$ for a of non-C type, and hence the weight J' rather than K_2'. Without the diagonal condition, a non-C type would be bounded only through P_b (taking b to be another element of the same type), giving the weight K_2' of Lemma 4.2; since $K_2'\approx 0.0501139 > 1/50$, the maximization of Section 6 would then collapse, and the $1/50$ bound could not be obtained by this method. The off-diagonal variant of the problem is therefore not addressed here.

6 The maximization

Set

$$F(S) = \frac{1}{M} \sum_{s \in S} f(s, S),$$

and let Σ be the family of all $S \subseteq \mathbb{Z}/M\mathbb{Z}$ that are not contained in the 7-class and not contained in the 18-class (i.e. $\{s \in S : s \not\equiv 7\,(25)\} \neq \emptyset$ and $\{s \in S : s \not\equiv 18\,(25)\} \neq \emptyset$). We must compute $\max_{S \in \Sigma} F(S)$.

6.1 Signature reduction

For $s \in \mathbb{Z}/M\mathbb{Z}$ define the *signature* $\sigma(s) = (s \bmod 4,\ s \bmod 9) \in \mathbb{Z}/4 \times \mathbb{Z}/9$; there are 36 signatures, each containing 25 types, of which 2 lie in C and 23 do not. Thus C has 72 elements, parametrized by (class $\in \{7, 18\}$, signature), and the complement of C consists of 36 groups of 23 types.

By (F3), gluing within a mod-25 class is automatic, while gluing between the two classes, and between a C-type and a non-C type, can only occur through 4 or 9 and therefore depends only on signatures: define the symmetric relation

$$(a, b)\ R\ (a', b') \quad \Longleftrightarrow \quad 4 \mid aa' + 1 \ \text{ or } \ 9 \mid bb' + 1.$$

That R computes gluing correctly in all the relevant cases was verified exhaustively over all 900×900 type pairs. Moreover all 23 non-C types of a given signature behave identically towards every C-type (verified exhaustively), and each carries the same weight J'; so a configuration is determined, for the purposes of F, by

$$(G_7,\ G_{18},\ Y) \subseteq (\mathbb{Z}/4 \times \mathbb{Z}/9)^3,$$

the signature sets of the 7-class types present, of the 18-class types present, and of the non-C groups present. For a signature set X write $R(X) = \bigcap_{i \in X} R(i)$ (with $R(\emptyset)$ the full set of 36 signatures). A signature set is *Galois-closed* if it is of the form $R(X)$, equivalently $R(R(X)) = X$.

Lemma 6.1 (Closure reduction)**.** *The maximum of F over Σ is attained at a configuration whose governing signature sets are Galois-closed. There are exactly* 64 *Galois-closed signature sets, and exactly* 441 *ordered pairs (A, B) of them satisfying the compatibility constraint $R(A) \subseteq B$; excluding the two pairs in which one side is empty and the other is the full set of* 36 *signatures, which correspond to configurations contained in a single extremal class and so lie outside Σ, leaves* 439 *pairs.*

Proof. Let (G_7, G_{18}, Y) be a configuration in Σ, so $G_{18} \cup Y$ and $G_7 \cup Y$ are both nonempty. Write $W_7 = G_7 \cap R(G_{18} \cup Y)$ and $W_{18} = G_{18} \cap R(G_7 \cup Y)$ for the signature sets receiving weight 1 on each side, and set $A = R(W_{18})$ and $B = R(W_7)$, which are Galois-closed. Every $i \in G_7$ is R-related to each $w \in W_{18} \subseteq G_{18}$, since $w \in R(G_7 \cup Y)$ means w is glued to every signature in $G_7 \cup Y$; by symmetry of R this gives $G_7 \subseteq A$, and the same argument gives $G_{18} \subseteq B$ and $Y \subseteq A \cap B$. Antitonicity of R then yields $R(A) \subseteq R(G_7) \subseteq R(W_7) = B$, so (A, B) is a compatible pair of closed sets, and $A \cap B$ is closed as an intersection of closed sets. In the replacement configuration $(A, B, A \cap B)$ the weight-1 set on the 7-side is $A \cap R(B \cup (A \cap B)) = A \cap R(B) \supseteq W_7$, because $R(B) = R(R(W_7)) \supseteq W_7$ and $W_7 \subseteq G_7 \subseteq A$; likewise on the 18-side. Hence the replacement enlarges G_7, G_{18}, Y and both weight-1 sets, and every added signature carries positive weight ($K_2' > 0$ for class signatures, $23J' > 0$ for non-C signatures), so F does not decrease. Therefore the maximum of F over Σ is attained at a configuration whose governing signature sets are Galois-closed. □

Remark 6.2 (Explicit list of the Galois-closed sets). The 64 closed sets can be written down by hand, so that the enumeration in Lemma 6.1 is checkable in principle without a computer. Identify the 36 signatures with pairs $(u, v) \in (\mathbb{Z}/4) \times (\mathbb{Z}/9)$ as above. Compatibility of two signatures is the *disjunction* of the mod-4 and the mod-9 condition, so for a single signature $i = (a, b)$,

$$R(i) = \big(R_4(a) \times \mathbb{Z}/9\big) \ \cup \ \big(\mathbb{Z}/4 \times R_9(b)\big),$$

where $R_4(a) = \{u : 4 \mid ua + 1\}$ and $R_9(b) = \{v : 9 \mid vb + 1\}$; these are singletons when a is odd, resp. when $3 \nmid b$, and empty otherwise. Intersecting such sets, one finds that the Galois-closed signature sets are exactly:

1. the empty set (1 set);
2. the full set of all 36 signatures (1);
3. the singletons $\{(u, v)\}$ with $u \in \{1, 3\}$ and v a unit mod 9 (12);
4. their duals $R(\{(u, v)\})$, each of size 12 (12);
5. the two-element sets $\{(1, v_1), (3, v_2)\}$ with $v_1 \neq v_2$ units mod 9 (30);
6. the sets $(\mathbb{Z}/4) \times \{v\}$, of size 4, with v a unit mod 9 (6);
7. the sets $\{u\} \times (\mathbb{Z}/9)$, of size 9, with $u \in \{1, 3\}$ (2).

Families (1), (2), (5), (6), (7) are preserved by R (as families), while R exchanges families (3) and (4). The total is $1+1+12+12+30+6+2 = 64$, in agreement with the enumeration.

6.2 The computation and its outcome

Theorem 6.3 (Computational)**.**

$$\max_{S \in \Sigma} F(S) = \frac{18}{900} = \frac{1}{50},$$

and the maximum is attained by exactly two configurations, namely

$$S_1 = \{s : \ s \bmod 100 \in \{7, 93\}\}, \qquad S_2 = \{s : \ s \bmod 100 \in \{43, 57\}\},$$

each consisting of 18 *types, all of weight* $f(s, S_i) = 1$*. Among the Galois-closed configurations, the second largest value of* F *is attained at* $S = \mathbb{Z}/900\mathbb{Z}$ *(all types present) and equals*

$$\frac{72K_2' + 828J'}{900} = \frac{3.60820 + 11.79484}{900} = \frac{15.40304}{900} \approx 0.0171145 < \frac{1}{50};$$

this is also the largest value among configurations containing at least one type outside C*.*

All decimal values quoted in this section are rounded for display; the underlying computations use the unrounded constants (e.g. $J' = 0.0142449748\ldots$, not the rounded 0.0142450). The rounding affects only the fifth decimal place of the products above and none of the inequalities used. Both extremal values in Theorem 6.3 are values of the upper-bound functional F; in particular the runner-up value is not the density of an admissible set, only the value of F at the all-types configuration (compare Section 8, where the largest admissible sets using types outside C have about $0.0155N$ elements in the computed range).

The optimum was computed in three ways, one by the author and two by an independent referee:

1. a CBC 0–1 integer program on the reduced model (72 binary variables for C-types, 36 for the non-C signature groups, plus full-weight indicator variables), by the author (`typebound.py`);

2. an exhaustive enumeration over the 64 Galois-closed signature sets (439 valid pairs), with each candidate realized as an explicit type set and F evaluated directly from its definition on all 900 types (`opt_closure.py`, independent referee);

3. a HiGHS MILP on the reduced model with no-good cuts, enumerating *all* optimal solutions and the next value below them (`opt_milp_reduced.py`, independent referee).

No brute-force search over subsets of $\mathbb{Z}/900\mathbb{Z}$ was performed, and none is needed. As a cross-validation, 3000 random admissible configurations S were evaluated directly and none exceeded $18/900$.

Remark 6.4 (The binding inequality and its robustness). Theorem 6.3 reduces the whole stability statement to the single numerical inequality

$$72K_2' + 828J' \;<\; 18,$$

which holds with room to spare: $72K_2' + 828J' = 15.40304$. At the true value of K_2' the inequality would still hold for any $J' < 0.01738$. It is, however, genuinely necessary to use the refined constant J' rather than K_1: replacing J' by K_1 gives $(72K_2' + 828K_1)/900 \approx 0.0292 > 1/50$.

7 Proof of Theorem 1.4

Proof of Theorem 1.4. Let $\varepsilon > 0$. Let $A \subseteq [N]$ be admissible, contained in neither extremal class, and let $S \subseteq \mathbb{Z}/900\mathbb{Z}$ be its set of types. If some $a \in A$ has $a \not\equiv 7 \ (25)$ then its type s satisfies $s \not\equiv 7 \ (25)$, and similarly for 18; hence $S \in \Sigma$.

By Proposition 5.1 (applied with $\varepsilon/2$) and Theorem 6.3,

$$|A| \;\le\; N\,F(S) + \frac{\varepsilon}{2}N + O(900) \;\le\; \frac{N}{50} + \frac{\varepsilon}{2}N + O(900),$$

which is at most $(1/50 + \varepsilon)N$ once $N \ge N_0(\varepsilon)$ with N_0 large enough to absorb the $O(900)$ term. This proves the upper bound.

Sharpness is Proposition 2.1: A_1 and A_2 are admissible, meet both classes, and have $N/50 + O(1)$ elements. That these are, within the mod-900 type framework, the only configurations attaining the bound is the uniqueness statement of Theorem 6.3, together with Proposition 2.2. □

Corollary 7.1 (Second-level stability)**.** *For every $\varepsilon > 0$ there is $N_0(\varepsilon)$ such that for all $N \ge N_0(\varepsilon)$, if $A \subseteq [N]$ is admissible, contained in neither extremal class, and*

$$|A| \;\ge\; \Big(\frac{72K_2' + 828J'}{900} + \varepsilon\Big)N \;\approx\; (0.0171145 + \varepsilon)N,$$

then $A \subseteq A_1$ or $A \subseteq A_2$: a mixed admissible set above the runner-up density is contained in one of the two extremal configurations.

Proof. Let S be the type set of A, with configuration (G_7, G_{18}, Y). By Proposition 5.1 applied with $\varepsilon/2$, $N\,F(S) \ge |A| - \frac{\varepsilon}{2}N - O(900)$, so $F(S) > (72K_2' + 828J')/900$ for N large. Apply the replacement of Lemma 6.1: the closed configuration $\big(R(W_{18}),\, R(W_7),\, R(W_{18}) \cap R(W_7)\big)$ has F at least that of S, and its realized type set contains S. By Theorem 6.3, the only

closed configurations in Σ with F above the runner-up value $(72K_2' + 828J')/900$ are the two maximizers S_1 and S_2 (the enumeration has no value strictly between the runner-up and 18/900). Hence $S \subseteq S_1$ or $S \subseteq S_2$, and reading residues modulo 100 gives $A \subseteq A_1$ or $A \subseteq A_2$. □

8 Numerical evidence for small N

Exact maximum-clique computations in the compatibility graph on $\{n \le N : n^2+1 \text{ not squarefree}\}$ give the following (these are consistent with, but of course weak evidence for, an asymptotic statement):

- For every $N \le 10000$, the overall maximum admissible size equals $|\{n \le N : n \equiv 7\ (25)\}|$, in agreement with Theorem 1.2.
- For every N with $557 \le N \le 10000$, the largest admissible set not contained in a single mod-25 class is exactly the $\{7, 93\}$ (or $\{43, 57\}$) mod 100 set. For some $N \le 556$ it is one element larger.
- The largest admissible set containing an element whose type lies outside C has about $0.0155N$ elements for $N = 4000$–10000, below the asymptotic runner-up value 0.0171145 of Theorem 6.3.

9 Computational components and verification status

We separate clearly what is computed, what is verified, and what is quoted as context.

(A) Finite computations entering the proof of Theorem 1.4.

1. *Local and gluing checks mod* 900. The facts (F1)–(F3) of Lemma 3.1, the signature reduction data (that all 23 non-C types of a signature behave identically towards each C-type, and that the relation R reproduces gluing in all cross-class and C/non-C cases), and Proposition 2.2, were verified by direct enumeration: (F1) for all primes $p < 2000$, (F2) on 20,000 random pairs, (F3) and the signature data on all 900×900 type pairs, and the mod-100 cross-class pairs exhaustively (and against all integer pairs up to 20000).
2. *The mod-*900 *maximization.* Theorem 6.3 is a finite computation, performed by the three programs listed in Section 6: the author's CBC 0–1 program, and two programs written separately by an independent referee (the exhaustive enumeration over the 64 Galois-closed candidate configurations, which relies on the closure reduction of Lemma 6.1, and a HiGHS MILP solve). The author's program shares no code with the referee's; all three agree on the optimal value and on the extremal configurations.

(B) The verified numerical range. A separate computation certifies the exact extremal bound of Theorem 1.2 in a finite range, by an explicit "seating-chart" (colouring) certificate. The generator (`gen848.c`) emits, for each plateau of the counting function $N \mapsto \lfloor (N+18)/25 \rfloor$, the plateau endpoints and the associated witness data; an independently written checker (`check848.c`, written from the format specification alone and using a different squarefree test) re-reads this output and verifies each plateau, so that an error in the generator cannot be absorbed silently. The conclusion certified is: for every

$$N \le 9{,}999{,}981,$$

comprising 399,999 plateaus, every admissible $A \subseteq [1, N]$ satisfies $|A| \le \#\{n \le N : \ n \equiv 7 \ (25)\}$. The certificate files have SHA-256 checksums

```
global: caa830054e120c7743031f8a1a3ec299365638f7
        a51b09a700baa9b5e54d94a1,
deltas: e78da84a7cbe1469959bca822738c27581a732ed8a8bf4
        df650a088f382a39d8,
```

(MD5: `e5b8573a1f1a0bc8bfe2a24651178195` and `e3330948328e523d101cd715af0f6a0a` respectively; the archive `certs_1e7.tgz` containing them has SHA-256 `680c3faa17589be8ba2772aa848103f1c4`. Both programs are distributed with the code (see (D) below). We stress that this computation does not enter the proof of Theorem 1.4: it corroborates the exact value $f(N)$ of Theorem 1.2 in the accessible range, independently of Li's formalization.

(C) Independent rebuild of Li's formalization. Theorem 1.2 relies on the Lean formalization of Li [2, 3]. We did not take the published build artefacts on trust. Instead we rebuilt the project — the 30,636 `Erdos848` modules behind the final theorem compiled from source, with Mathlib and its dependencies taken from the official Mathlib build cache — and replayed the kernel checks. Three points are worth recording.

1. *Patch.* The repository as published builds with Li's own build driver, which compiles each file with `lake env lean` (where auto-bound implicit variables are allowed). A plain `lake build`, however, applies the project's `autoImplicit = false` setting, and two declarations rely on auto-bound implicit variables; under that setting the module stalls (the follow-on elaboration failure does not terminate, as that file sets `maxHeartbeats 0`); a two-line patch adds the binders $\{v\ w : \mathbb{N}\}$ and $\{w : \mathbb{N}\}$, with no change to any statement (the elaborated types are identical, checked with `pp.all`). The patch is included verbatim in our repository so that the discrepancy between the published source and the compiling source is visible rather than hidden.

2. *Build.* With this patch the full project builds cleanly: 39,069 jobs, exit code 0, no `sorry` and no errors. (Two `Publication*` files in the repository are not part of the build for the final theorem.)

3. *Kernel replay and axiom audit.* We ran `#print axioms` on the top-level statement `all_N` and obtained exactly

   ```
   [propext, Classical.choice, Quot.sound],
   ```

   i.e. the three standard axioms of Lean itself and nothing further — in particular no `sorryAx` and no unverified native-code evaluation axiom. Independently, a run of `leanchecker` over the compiled environment re-checked the 30,636 project modules with 0 failures (Mathlib itself was not replayed), so the proof terms were re-run through the Lean kernel rather than merely accepted from the compiler's cache.

Thus the theorem quoted from [2, 3] is a machine-checked theorem whose dependency set we have audited, not an informal assertion.

(D) Availability. All code described above is publicly available at the address below: in `stability/`, the author's and the referee's density computations, the referee's checks of (F1)–(F3), of the signature reduction and of the worst-case argument (`local_facts.py`,

`closure_crosscheck.py`, `worstcase_local.py`, `consts.py`), the three mod-900 solvers and the small-N clique computations; in `cert/`, the certificate generator and the independent certificate checker; and in `li-replay/`, the binder patch and the build, statement-check and kernel-replay scripts and logs for (C):

https://github.com/Paulee314/erdos848

together with the raw output files whose checksums are quoted in (B).

Summary of logical status. Theorem 1.4 depends on: (i) the analytic density bounds of Section 4, proved here from first principles; and (ii) the finite checks (A), each of which is an exhaustive enumeration over an explicitly bounded search space. It does *not* depend on the formalized result of Li [2, 3], and it does *not* depend on the stability statement of Sawhney [4]. Theorem 1.2 is quoted only for the exact extremal value $f(N) = \lfloor (N+18)/25 \rfloor$ attained inside a single extremal class, which is what allows the picture to be described as complete; the audit (C) is reported for readers who wish to rely on that value, not because Theorem 1.4 rests on it. Theorem 1.3 is the predecessor statement that the present work sharpens: Theorem 1.4 implies Theorem 1.3 for every fixed $\eta < 1/50$ (take $\varepsilon < 1/50 - \eta$ in Theorem 1.4), so the admissible η is improved from ≈ 0.0023 to anything below $1/50 = 0.02$. Finally, Theorem 1.4 does not depend on the numerical range (B), which is reported as independent corroborating evidence.

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